Automatic Temperature Controlled Fan using Arduino Uno (Tinkercad Simulation)

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Tools used: Tinkercad Circuits, Arduino Uno, TMP36 temperature sensor, NPN transistor, DC motor (fan), 9V Battery, diode

1. Abstract

This project presents the design and simulation of an Automatic Temperature Controlled Fan system using an Arduino Uno on the Tinkercad Circuits platform. The system monitors ambient temperature through a TMP36 sensor and automatically switches a DC fan ON or OFF to maintain desired thermal limits without manual intervention.

The analog voltage from the TMP36 sensor is read by the Arduino Uno on pin A0 and converted into degrees Celsius in software. When the temperature reaches 30 °C, the Arduino drives a DC motor (acting as the fan) through an NPN transistor. Below this limit the fan stays OFF. The live temperature is also shown on the Serial Monitor, and the design is verified through simulation.

Keywords: Arduino Uno, Tinkercad Circuits, TMP36, NPN Transistor, DC Motor, Serial Monitor, Temperature Control

2. Introduction

In modern electronic systems and automated environments, temperature regulation is crucial. Traditional manual switching of cooling systems is inefficient. This project aims to build an automated solution that detects temperature changes and switches a DC fan ON or OFF dynamically, saving energy and preventing overheating.

Excess heat is a common cause of failure in electronic equipment, and a fan that runs continuously wastes power and makes unnecessary noise. A temperature controlled fan avoids this by running only when it is actually needed.

Embedded systems are well suited to this kind of control task. A microcontroller can read a sensor, compare the reading with a set value and switch an output device within a fraction of a second, all without human involvement. In this project, the Arduino Uno acts as the brain of the

system. The TMP36 sensor gives it an analog voltage that depends on the surrounding temperature. The Arduino converts this voltage into degrees Celsius and compares it with a threshold of 30 °C. When the threshold is reached, it switches the DC motor (fan) ON through an NPN transistor, and when the temperature falls below the threshold, it switches the fan OFF.

2.1 Objectives

• To sense ambient temperature continuously using the TMP36 sensor.

• To switch a DC fan ON or OFF automatically through an NPN transistor at the 30 °C threshold.

• To display the current temperature on the Serial Monitor for observation.

• To verify the complete design through simulation in Tinkercad Circuits.

2.2 Key Highlights

• Automatic operation: The fan turns ON and OFF by itself based on temperature

• Energy saving: The fan runs only when cooling is actually needed

• Live monitoring: Temperature is printed on the Serial Monitor every second

• Simulation based: Complete circuit and code verified in Tinkercad Circuits

3. System Architecture and Components Used

The system consists of three main units that automatically control a cooling fan based on temperature:

TMP36 Sensor (Analog Pin A0) - rightarrow Arduino Uno (Digital Pin 3) - rightarrow NPN Transistor - rightarrow DC Motor (Fan)

3.1 Block Diagram Components

• Input Unit (TMP36 Sensor): Measures ambient temperature and sends a proportional analog voltage to Arduino pin A0 (0–1023 digital value).

• Processing Unit (Arduino Uno): Converts the sensor reading into degrees Celsius, compares it against a 30 °C threshold, and controls pin 3 accordingly.

• Output Unit (NPN Transistor & DC Motor): Uses an NPN transistor as a current amplifier/switch to turn the DC cooling fan ON (pin HIGH) or OFF (pin LOW).

3.2 Software Requirements

• Tinkercad Circuits: Building the circuit and running the real-time simulation

• Tinkercad Code Editor: Writing the Arduino program (C++) and uploading it to the virtual Arduino Uno

• Serial Monitor: Viewing the live temperature readings at 9600 baud

4. Circuit Diagram

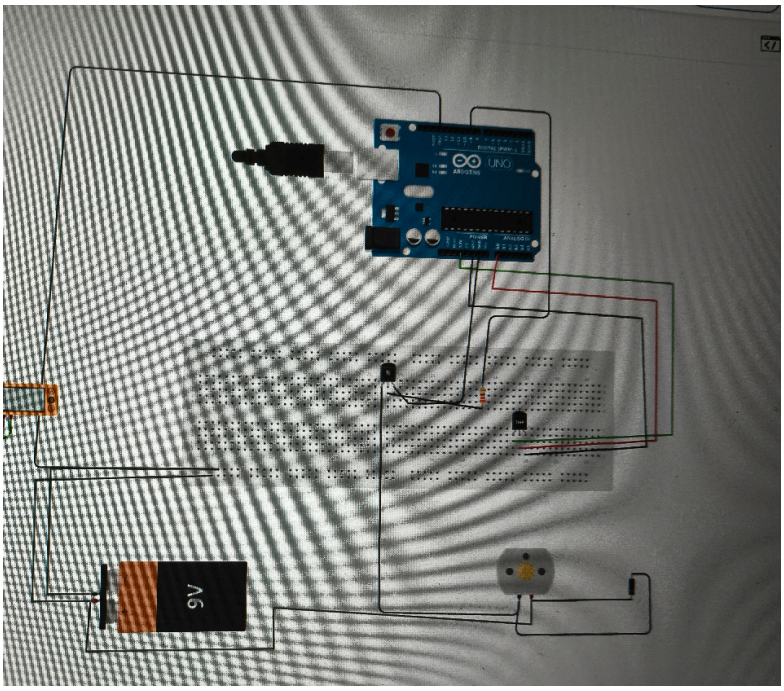


Fig: Tinkercad Circuit Diagram of the Automatic Temperature Controlled Fan

4.1 Connection Description

The TMP36 sensor output is connected to analog input pin A0 of the Arduino Uno. Digital output pin 3 of the Arduino controls the NPN transistor, which switches the DC motor (fan) ON or OFF based on the temperature decision taken by the program.

4.2 Pin Connection Summary

• TMP36 output to Arduino pin A0: Analog temperature input

• Arduino pin 3 to NPN transistor (control input): Fan ON / OFF control

• Transistor output to DC motor: Acts as the cooling fan

5. Source Code

```
const int tempPin = A0;   // TMP36 Sensor Analog Input Pin
const int fanPin  = 3;    // DC Motor Control Digital Output Pin

void setup() {
   pinMode(fanPin, OUTPUT);  // Set fanPin as output
   pinMode(tempPin, INPUT);  // Set tempPin as input
   Serial.begin(9600);       // Initialize Serial Monitor at 9600 baud
}

void loop() {
   // Read raw analog value from TMP36 sensor (0 to 1023)
   int sensorValue = analogRead(tempPin);

   // Convert analog reading to voltage (0V to 5V)
   float voltage = sensorValue * (5.0 / 1023.0);

   // Convert voltage to Celsius temperature (°C)
   float temperatureC = (voltage - 0.5) * 100.0;

   // Print temperature data to Serial Monitor
   Serial.print("Current Temperature: ");
   Serial.print(temperatureC);
   Serial.println(" deg C");

   // Threshold Control Logic
   if (temperatureC >= 30.0) {
      digitalWrite(fanPin, HIGH);   // Turn Fan ON
   } else {
      digitalWrite(fanPin, LOW);    // Turn Fan OFF
   }
   delay(1000);   // Sampling delay of 1 second
}
```

5.1 Code Explanation

• Constants (tempPin, fanPin): tempPin = A0 is the TMP36 analog input; fanPin = 3 is the digital output that controls the DC motor

• setup(): Sets fanPin as OUTPUT, tempPin as INPUT and starts the Serial Monitor at 9600 baud

• analogRead(): Reads the raw sensor value from 0 to 1023 (10-bit ADC for 0 V to 5 V)

• Voltage and temperature: voltage = value x (5.0 / 1023.0); temperature = (voltage - 0.5) x 100, as the TMP36 gives 10 mV per °C with 0.5 V offset

• Threshold logic: If temperature >= 30.0 °C, pin 3 goes HIGH (fan ON); otherwise LOW (fan OFF)

• Serial output and delay(1000): Prints the temperature in °C on the Serial Monitor, and the sensor is sampled once every 1 second

6. Working Principle, Simulation Output and Conclusion

6.1 Working Principle

The TMP36 sensor measures the temperature and sends an analog voltage to pin A0. The Arduino Uno converts it into degrees Celsius every second. When the temperature reaches 30 °C, pin 3 goes HIGH, the NPN transistor conducts and the fan starts. When it falls below 30 °C, the pin goes LOW and the fan turns off.

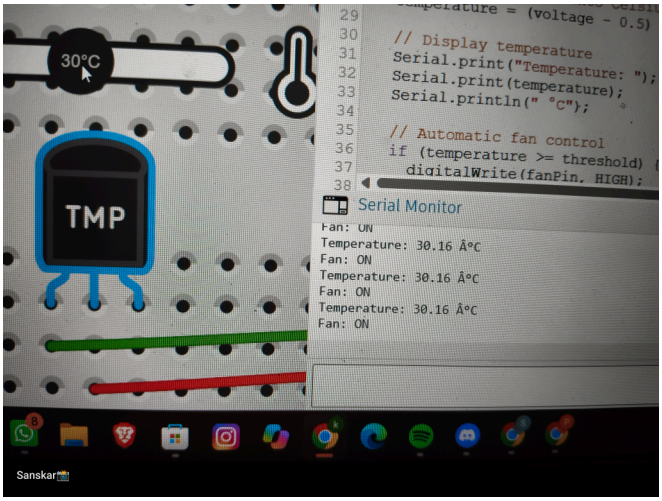
• Sensor value 154: Voltage 0.75 V, Temperature 25.3 °C, Fan OFF

• Sensor value 164: Voltage 0.80 V, Temperature 30.2 °C, Fan ON

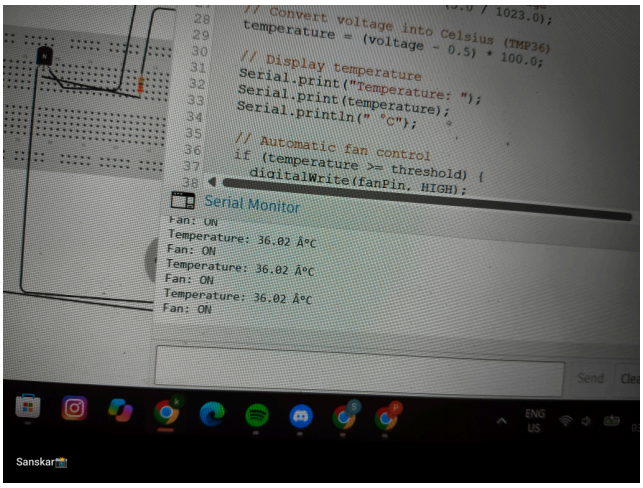
• Sensor value 175: Voltage 0.86 V, Temperature 35.5 °C, Fan ON

6.2 Simulation Output

1. If temperature is 30°C



2. If temperature is greater than 30°C



3.If temperature is less than 30°C

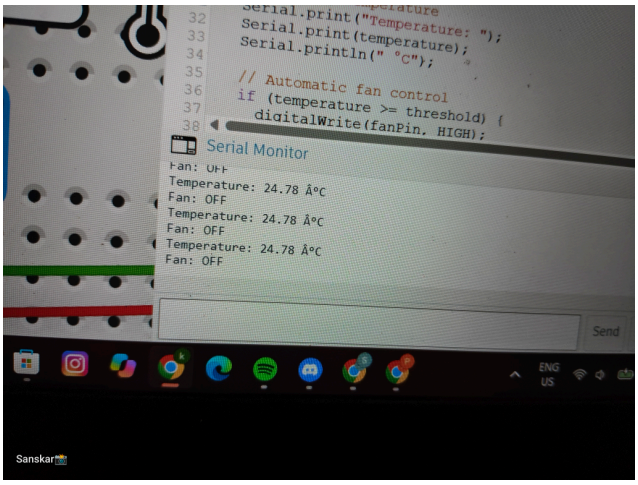


Fig : Tinkercad Simulation Output (Serial Monitor and Fan Status)

6.3 Real-World Applications

• Server rooms cooling

• Industrial machinery thermal protection

• Automated greenhouse ventilation

6.4 Conclusion

The Automatic Temperature Controlled Fan was successfully simulated using an Arduino Uno, a TMP36 sensor and a DC motor driven through an NPN transistor. The Serial Monitor readings confirmed that the fan stays OFF at 24.78 °C and switches ON at 30.16 °C and above, once the 30 °C threshold is reached. The system controls the fan automatically and saves energy by running it only when cooling is needed.